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Testing for unit roots in seasonal time series

Some theoretical extensions and a Monte Carlo investigation*

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Part of the increasing interest in the treatment of seasonality in economic time series has focused on detecting the presence of unit roots at some of the seasonal frequencies as well as at the zero frequency. In this paper we introduce new test statistics, analyze both theoretically and via simulation the properties of Dickey–Fuller-type tests in seasonal time series which have roots at frequencies other than the zero frequency. We also investigate the properties of the standard testing procedures for unit roots in seasonal time series via Monte Carlo simulations. We show that the Dickey–Fuller tests can still be used to test for a unit root at the zero frequency to the extent that appropriate autoregressive correction terms are augmented to the model. Our Monte Carlo simulations reveal that tests for unit roots at seasonal frequencies have severe size distortions in many cases commonly encountered in practice. While we find the procedure proposed by Hylleberg, Engle, Granger, and Yoo (1990) the most useful among the alternative procedures, we caution users of many remaining serious obstacles when testing for unit roots in seasonal time series.

Key words: HEGY procedure; Dickey–Fuller tests
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1. Introduction

The purpose of our paper is twofold. First, we investigate, via Monte Carlo simulations, the properties of some standard testing procedures for unit roots in seasonal time series proposed by Dickey, Hasza, and Fuller [henceforth, DHF] (1984) and Hylleberg, Engle, Granger, and Yoo [henceforth, HEGY] (1990).[1] Second, we examine the effect of seasonal unit roots on the usual tests for a unit root at the zero frequency, and analyze the behavior of the Dickey–Fuller [henceforth, DF] (1979) test in cases where time series processes contain unit roots at some seasonal frequencies as well as at the zero frequency. We study existing procedures for unit roots and seasonal unit roots and also introduce new test statistics. One advantage of the HEGY procedure over that proposed by DHF is that one can test for unit roots at each frequency separately without maintaining that there are unit roots at some or all of the other frequencies. In order to compare the DHF and HEGY procedures directly, we introduce a joint HEGY-type test for the presence of unit roots at the zero and all seasonal frequencies.

Our Monte Carlo experiments suggest that the HEGY testing procedure has some clear advantages because of its design. Indeed, when the data-generating processes contain a unit root only at the zero frequency or at one of the seasonal frequencies, the DHF procedure appears unable to separate this unit root with nonstationarity induced by the $(1 - B^4)$ factor. In general, however, our Monte Carlo study reveals that there are some serious problems with currently available tests for unit roots at seasonal frequencies. Near-cancellations of a unit root in the AR polynomial with a MA root are known to adversely affect size properties of unit root tests [see e.g., Schwert (1989)]. In seasonal time series models such near-cancellations are quite common for seasonal unit roots. It is noted, in fact, that size distortions are very serious in many practical circumstances. Moreover, in cases where there are no size distortions, it is also found that neither the DHF nor the HEGY tests have good power properties against deterministic seasonal component alternatives, particularly when seasonal dummies are excluded from the regression models.

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[1] We restricted our attention to those tests as they appear to be the most often encountered in the literature. However, a number of other tests for nonstationarity in seasonal time series have been used: notably, tests by Hasza and Fuller (1982) and Osborn et al. (1988) for multiplicative models and tests based on vectors of stacked seasonal processes as in Franses (1990) allowing to study periodic models [see also Osborn (1991)].

A direct relationship between the standard DF t-statistic and the HEGY test for the zero frequency unit root is also established. While their asymptotic equivalence is discussed in HEGY, it is shown here that this equivalence extends to finite samples. Moreover, we also demonstrate that using DF t-statistics remains valid even when unit roots are present at frequencies other than zero, though special care must be taken regarding the AR polynomial expansion. This has important practical implications, as one can continue to apply DF t-statistics in seasonal time series, regardless of their stochastic seasonal characterization. Finally, it is also shown that the finite sample and asymptotic equivalence results between DF and HEGY do not directly carry over to a normalized bias test statistic.

In section 2, we introduce the alternative testing procedures for unit roots and seasonal unit roots employed in this paper. We also discuss the properties of the usual DF test for a unit root when time series processes contain unit roots at some frequencies other than zero. Section 3 presents results of Monte Carlo experiments that calculate the size and power of the alternative testing procedures for a variety of data-generating processes. Section 4 concludes with a brief summary of the simulation results and suggestions for further research.

2. Testing procedures

In a first subsection 2.1, we briefly discuss the standard procedures as they were proposed by DHF and HEGY. Some thoeretical extensions of the HEGY procedure are then introduced in subsection 2.2. Finally, we discuss the properties of the DF procedures in seasonal time series in subsection 2.3.

2.1. Some standard testing procedures – A review

Throughout this paper, we consider the quarterly case. Let x_t be a univariate stochastic process:

$$x_t = \alpha x_{t-4} + u_t, \qquad t = 1, 2, \ldots, T, \tag{2.1}$$

where u_t is either a stationary process with zero mean and constant variance or, alternatively, depending on the context and test, u_t is a martingale difference sequence following the regularity conditions appearing in Phillips (1987) and Chan and Wei (1988). The x_t process has four roots on the unit circle when $\alpha = 1$; one at frequency $\theta = 0$, one at frequency $\theta = \pi$, and a pair of complex roots at frequencies $\theta = \pi/2$, $3\pi/2$. The framework proposed by DHF is discussed first and is then followed by that of HEGY.

2.1.1. The DHF tests

DHF (1984) examined several regression-based tests for the hypothesis that $\alpha = 1$ in model (2.1) where u_t is i.i.d. N(0, 1) and computed percentiles of their distributions. They suggested two types of test statistics derived from the least squares estimator in regression models with and without a constant and/or seasonal dummies. Following the DHF notation, let $\hat{\alpha}_4$, $\hat{\alpha}^*_{\mu 4}$, and $\hat{\alpha}_{\mu 4}$ be the least squares estimator from the regression of x_t on x_{t-4} with no deterministic terms, with a constant only and with four seasonal dummies, respectively. We will examine the normalized biases, i.e., $T(\hat{\alpha}_4 - 1)$, $T(\hat{\alpha}^*_{\mu 4} - 1)$, and $T(\hat{\alpha}_{\mu 4} - 1)$, and the studentized regression statistics, τ_4, $\tau^*_{\mu 4}$, and $\tau_{\mu 4}$. The limiting distributions of these test statistics and their critical values appeared in the original paper. As is the case with DF tests, we can accommodate serial correlation in u_t by augmenting the auxiliary regression with an AR polynomial in $x_t - x_{t-4}$. In section 3, we will describe in greater detail the practical implementation of DHF tests.

2.1.2. The HEGY tests

HEGY (1990) introduced a factorization of the seasonal differencing polynomial $\Delta_4 \equiv (1 - B^4)$ and developed a testing procedure for seasonal unit roots, which consists in estimating via OLS the following regression:

$$\Delta_4 x_t = \pi_1 y_{1,t-1} + \pi_2 y_{2,t-1} + \pi_3 y_{3,t-2} + \pi_4 y_{3,t-1} + \varepsilon_t , \tag{2.2}$$

where $y_{1t} = (1 + B + B^2 + B^3)x_t$, $y_{2t} = -(1 - B + B^2 - B^3)x_t$, and $y_{3t} = -(1 - B^2)x_t$. Notice that when $\alpha = 1$ in (2.1), y_{1t}, y_{2t}, and y_{3t} have unit roots only at $\theta = 0$, π, and $\pi/2$, respectively. Hence, the unit root found at $\theta = 0$ in x_t implies accepting the null that π_1 is zero. Similarly, when π_2 is zero, it indicates the existence of a unit root at $\theta = \pi$. When both π_3 and π_4 are zero, then we have a pair of complex unit roots at $\theta = \pi/2$. HEGY proposed t-statistics for π_1, π_2, and π_3 if ($\pi_4 = 0$), which will be denoted t_1 through t_4, as well as F-tests for π_3 and π_4 jointly denoted F_{34}. Similar to DHF, each test was considered for regression models with and without a constant, trend, or seasonal dummies. Likewise, one can expand the regression model via an AR polynomial augmentation. In section 3, we will again describe in greater detail the practical implementation of HEGY tests.

2.2. Extensions of the HEGY procedure

The DHF t-statistic for $\alpha = 1$ in (2.1) tests the null of unit roots at $\theta = 0$, $\pi/2$, and π simultaneously. A comparable HEGY-type test would amount to an

F-statistic on π_1, π_2, π_3, and π_4. The asymptotic distribution of such a test can be derived from the results appearing in Engle, Granger, Hylleberg, and Lee (1993). Specifically, in the HEGY regression (2.2), we have that

$$F_{1234} \to \frac{1}{4}\left\{\frac{(\int_0^1 W_1(r)\mathrm{d}W_1)^2}{\int_0^1 W_1(r)^2\,\mathrm{d}r} + \frac{(\int_0^1 W_2(r)\mathrm{d}W_2)^2}{\int_0^1 W_2(r)^2\,\mathrm{d}r} + \frac{(\int_0^1 W_3(r)\mathrm{d}W_3 + \int_0^1 W_4(r)\mathrm{d}W_4)^2 + (\int_0^1 W_3(r)\mathrm{d}W_4 - \int_0^1 W_4(r)\mathrm{d}W_3)^2}{\int_0^1 W_3(r)^2\mathrm{d}r + \int_0^1 W_4(r)^2\mathrm{d}r}\right\}, \tag{2.3}$$

where '$\to$' denotes weak convergence in distribution and $W_i(r)$ for $i = 1, \ldots, 4$ are the standard Brownian motions that are mutually independent. It is worth noting that the F-statistic in (2.3) has the same asymptotic distribution as the sum of the squared t-statistics for π_i ($i = 1, \ldots, 4$). A sketch of the proof appears in appendix A.[2]

Another useful extension of the HEGY procedure is an F-statistic for testing the presence of unit roots at all the seasonal frequencies simultaneously but not considering the zero frequency. The asymptotic distribution of the F-statistic for the null that $\pi_2 = \pi_3 = \pi_4 = 0$ is as follows:

$$F_{234} \to \frac{1}{3}\left\{\frac{(\int_0^1 W_2(r)\mathrm{d}W_2)^2}{\int_0^1 W_2(r)^2\,\mathrm{d}r} + \frac{(\int_0^1 W_3(r)\mathrm{d}W_3 + \int_0^1 W_4(r)\mathrm{d}W_4)^2 + (\int_0^1 W_3(r)\mathrm{d}W_4 - \int_0^1 W_4(r)\mathrm{d}W_3)^2}{\int_0^1 W_3(r)^2\mathrm{d}r + \int_0^1 W_4(r)^2\mathrm{d}r}\right\}.$$

The proof also appears in appendix A where it is shown that this test has the same limiting distribution as the sum of the corresponding squared t-statistics. We calculated via simulation the critical values for both F_{1234} and F_{234} for a number of sample sizes which appear in table C.1 of appendix C.

2.3. DF tests and seasonal time series

The commonly assumed regularity conditions for applying augmented Dickey–Fuller [henceforth, ADF] tests imply that the DGP does not have roots at frequencies other than the zero frequency and that there is at most one unit

[2] One should expect that the power properties of the F-statistic in (2.3) are different from those of the DHF t-statistics against some alternatives, as the former is based on the sum of squared t-statistics which can be viewed as a two-sided version of the DHF test.

root at the zero frequency. It is, therefore, worth investigating the properties of DF-type unit root tests when unit roots are present at some of the seasonal frequencies. We consider the standard DF regression models with and without a constant and trend. Moreover, we also consider regressions which include seasonal dummies. Following the results in Dickey, Bell, and Miller (1986), we can use the critical values of DF tests with a constant term when seasonal dummies appear in the regression equation.

To examine the properties of the DF test for a unit root at the zero frequency in the presence of seasonal unit roots, we can rewrite the time series process in (2.1) as follows:

$$\Delta x_t = \phi_1 x_{t-1} + \phi_2 \Delta x_{t-1} + \phi_3 \Delta x_{t-2} + \phi_4 \Delta x_{t-3} + u_t, \tag{2.4}$$

where $\phi_1 = \alpha - 1$ and $\phi_2 = \phi_3 = \phi_4 = -\alpha$. Assuming that u_t is a martingale difference sequence satisfying the conditions for a functional central limit theorem [see Phillips (1987)], we can show, using the results in Chan and Wei (1988), that under the hypothesis that $\alpha = 1$ (or equivalently $\phi_1 = 0$) the following relations hold:

$$(T/4)\hat{\phi}_1 \to \{W(1)^2 - 1\}/2\textstyle\int_0^1 W(r)\mathrm{d}r, \tag{2.5}$$

$$t_{\phi_1} \to \{W(1)^2 - 1\}/2[\textstyle\int_0^1 W(r)^2 \mathrm{d}r]^{1/2}, \tag{2.6}$$

where $W(r)$ denotes a standard Brownian motion on [0, 1].

The result in (2.6) shows that the usual Dickey–Fuller t-statistics can still be used to test the hypothesis of a unit root at the zero frequency, even in the presence of unit roots at other seasonal frequencies to the extent that lagged terms of dependent variables are appropriately augmented as in (2.4). In this case, the limiting distribution of the t-statistic in (2.6) is exactly the same as the one discussed in DF (1979) where the data-generating process was assumed to contain a unit root only at the zero frequency. The result in (2.5), however, suggests that the normalized-bias statistic is still applicable, but should be divided by four to have the same limiting distribution as that in DF (1979).

In addition to the validity of the DF tests in time series with unit roots at the seasonal frequencies, there is also an asymptotic equivalence between HEGY-type tests and DF procedures. Indeed, HEGY (1990) found that their statistic for π_1 has the same limiting distribution as the DF t-statistic. This asymptotic equivalence, in fact, carries over to finite samples, as is shown in appendix B.

As an aside, we can exploit both the finite sample and asymptotic equivalence between HEGY-type tests and the DF procedure to show that the latter can be used to test for the presence of unit roots at frequencies other than the zero frequency. Namely, if we decide to test for a seasonal unit root, say, at the biannual frequency, we can use the t-statistics on β_1 in the regression model:

$$(1+B)x_t = \beta_1(-x_{t-1}) + \beta_2(1+B)x_{t-1} + \beta_3(1+B)x_{t-2} + \beta_4(1+B)x_{t-3} + u_t .$$

From our previous discussion, we can show that the normalized-bias test statistic, $(T/4)(\hat{\beta}_1 - 0)$, and the t-statistic, t_{β_1}, have the same limiting distribution as in (2.5) and (2.6), respectively.

3. Monte Carlo experiments

In this section, we describe the design of our experiments in subsection 3.1. Next, we report and discuss the results in subsection 3.2. For the benefit of the reader, however, let us first briefly describe the regression models and tests. The DHF tests are based on the following regression:

$$x_t = \alpha x_{t-4} + \sum_{j=1}^{l} \Delta_4 x_{t-j} + [\text{set of fixed regressors}] + \varepsilon_t , \tag{3.1}$$

where the set of fixed regressors may include a constant, seasonal dummies, and a linear trend. Regression (3.1) is estimated with OLS. DHF proposed two types of statistics for H_0: $\alpha = 1$, based on $\hat{\alpha}$ in (3.1) – namely, a normalized statistic $T(\hat{\alpha} - 1)$ and a t-statistic. In appendix table C.2, we reproduced the critical values of the tests for the different configurations of fixed regressors and the sample sizes used in our Monte Carlo study. The HEGY tests involve running a different regression, namely,

$$\Delta_4 x_t = \pi_1 y_{1,t-1} + \pi_2 y_{2,t-1} + \pi_3 y_{3,t-2} + \pi_4 y_{3,t-1} + \sum_{j=1}^{l} \Delta_4 x_{t-j} + [\text{set of fixed regressors}] + \varepsilon_t , \tag{3.2}$$

where y_{it}, $i = 1,2,3$, are defined in (2.2). Eq. (3.2) involves the same set of fixed regressors as in (3.1). Again, using OLS, we can compute the t-statistics for $\pi_1 = 0$ (zero frequency unit root), denoted t_1, and for $\pi_2 = 0$ (unit root at π), denoted t_2. To test the presence of a pair of complex unit roots at $\pi/2$, one can use the t-tests for $\pi_3 = 0$, denoted t_3, and if the null hypothesis is accepted, compute the t-test for $\pi_4 = 0$, denoted t_4. Alternatively, one can perform joint F-tests for the pair of complex roots at $\pi/2$, i.e., $\pi_3 = \pi_4 = 0$. This test will be denoted F_{34}. Finally, the tests introduced in section 2.1, namely F_{1234} and F_{234}, are also F-tests computed from (3.2), respectively for $\pi_2 = \pi_3 = \pi_4 = 0$, i.e., no seasonal unit root and $\pi_1 = \pi_2 = \pi_3 = \pi_4 = 0$, the latter corresponding to the null hypothesis $\alpha = 1$ in the DHF regression (3.1). The critical values are reported in appendix tables C.1 and C.2. They were computed for each set of regressors. Finally, we also computed DF statistics based on

$$x_t = \alpha x_{t-1} + \sum_{j=1}^{l} \Delta x_{t-j} + [\text{set of fixed regressors}] + \varepsilon_t \,. \tag{3.3}$$

Here, the null of $\alpha = 1$ is of interest. With the same set of fixed regressors as before, one can compute a t-statistic based on $\hat{\alpha}$, the OLS estimator, for a zero frequency unit root.

3.1. Design of the data-generating processes

We considered three types of data-generating processes [henceforth, DGP]. The first one is a quarterly seasonal autoregressive moving average model possibly integrated of order 1, that we will denote as a SARIMA$(p, d, q)_4$ model. The second DGP is a SARIMA$(p, d, 0)_4$ augmented with deterministic seasonal dummies. Finally, our third and last class of DGP's consists of regular autoregressive models with deterministic seasonal components. We now describe the three DGP's in detail.

3.1.1. The SARIMA$(p, d, q)_4$ models

One of the common characterizations of stochastic seasonal time series is

$$x_t = \alpha x_{t-4} + u_t \,, \qquad t = 1, \dots, T \,, \tag{3.4}$$

where u_t is a stationary process. With $\alpha = 1$, we recover the null hypothesis of the various DHF statistics as well as the F_{1234}-statistic introduced in section 2.2. Moreover, with $\alpha = 1$ in (3.4), we have a unit root at all frequencies $\theta = 0$, $\pi/2$, and π, which correspond to the null hypotheses of t_1 through t_4 as well as F_{234} and F_{34}. Eq. (3.4) can be used to study the size of these tests when u_t is white noise or serially correlated via the following MA representation:

$$u_t = \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_4 \varepsilon_{t-4} \,. \tag{3.5}$$

As the testing procedures described in section 2 are based on autoregressive representations of time series data, the moving average component is approximated by a finite-order AR process. It has been shown that test statistics for a unit root in mixed autoregressive and moving average processes using high-order AR approximations have the same limiting distribution as the usual tests for a unit root in pure autoregressive processes [see Said and Dickey (1984)]. However, finite-order AR models used in practical applications may not provide adequate approximations to mixed AR and MA processes, as discussed, for instance, in Schwert (1989).

In light of the existing results in the literature, we took a restricted MA(4), as represented in (3.5), where θ_1 appears as the first lag coefficient and θ_4 as the one at lag 4. With those coefficients nonzero, we can study, for instance, near-cancellations of $(1 - L)$, $(1 + L)$, and $(1 - L^4)$ in the autoregressive part. The parameters considered to analyze the size of the tests described in the previous section are the following: $\theta_i = -0.9, -0.5, 0, 0.5, 0.9$ for $i = 1, 4$. In addition to size simulations, we also considered the power of each test with $\alpha = 0.85$.[3]

3.1.2. The SARIMA $(p, d, 0)_4$ models with seasonal dummies

The first DGP is purely stochastic. To document the effect of deterministic seasonal components on the size and power properties of tests, we also consider SARIMA $(p, d, 0)_4$ models with seasonal dummies described as

$$x_t = \alpha x_{t-4} + \sum_{i=1}^{4} \delta_i SD_{it} + u_t, \qquad t = 1, 2, \ldots, T, \tag{3.6}$$

where SD_{it} is a set of seasonal dummies $(i = 1, 2, 3, 4)$ and u_t is white noise. We consider two sets of parameter values for $(\delta_i)_{i=1,\ldots,4}$, namely, $(-0.1, 0.05, -0.05, 0.1)$ and $(-1, 0.5, -0.5, 1)$. The first parameter set can be interpreted as representing a seasonal pattern corresponding to a 10% (annualized) drop followed by a 5% increase, a 5% decrease, and finally a 10% rise. Such a pattern is fairly common for many seasonal time series. The second parameter set is just the tenfold of the first. Its design is meant to pick up effects of the magnitude of deterministic seasonal variation on the statistical properties of tests. It should also be noted that the deterministic seasonal variation sums to zero. Hence, there is no drift in the DGP. The parameter values for α are the same as those described in the first model specification.

3.1.3. Autoregressive models with seasonal dummies

The third and last specification is as follows:

$$x_t = \alpha x_{t-1} + \sum_{i=1}^{4} \delta_i SD_{it} + u_t, \qquad t = 1, 2, \ldots, T. \tag{3.7}$$

[3] Please note that α is the fourth lag coefficient so that $\alpha^{1/4} = 0.96$. The latter is probably a better figure to use when interpreting the results.

Hence, we consider a simple first-order autoregressive model with deterministic seasonal variation. The parameter values for $(\delta_i)_{i=1,\ldots,4}$ are the same as those previously described. In addition, we also considered the case $\delta_i = 0$ for all i, corresponding to a pure AR(1) model with no unit roots at the $\theta = \pi/2$ and π frequencies and possibly no zero frequency unit root when $\alpha \neq 1$. The parameter for $\alpha = 1$ and 0.9 are considered in this case to analyze size and power respectively.

3.2. *Monte Carlo results*

The size and power properties of the alternative testing procedures are investigated in this section via Monte Carlo experiments using the data-generating processes in (3.4), (3.6), and (3.7). Model specification sensitivity is analyzed using different numbers of lags, 0 up to 10, for the AR expansions and additional fixed regressors (i.e., a constant, seasonal dummies, and/or a linear trend) were included in the regression equations. Samples of size $T = 48$, 100, 160, 200, 400 are used in the experiments. To keep the volume of tables to a reasonable level we only report a selection of our simulation results. Specifically, the results with lag length $l = 0, 1, 3, 4, 8$ and with samples of $T = 100$, 400 only are selectively reported.[4] In addition, only the size and power estimates based on the 5% critical values are reported in the tables. The figures obtained using the 10% critical values, while available upon request, are omitted in this paper as they reveal essentially the same features. Moreover, we do not report results involving $\theta_i = -0.5$ or 0.5 in (3.5). Only the parameter configurations $\theta_i = -0.9, 0.0, 0.9$ appear in our tables. The size properties of the tests are discussed in the first subsection 3.2.1, the power of the tests follows in the next subsection.

3.2.1. *Size of the tests*

The effects of moving-average components on the size of the testing procedures for unit roots based on autoregressive expansions are first examined using the DGP (3.4) with $\alpha = 1$. In this case, the simulation results will show the influence of model misspecification on the size of the tests. While the presence of high MA components and their near-cancellation effect may be more an issue of theoretical relevance for zero frequency unit root tests, it is of much more practical relevance in seasonal models. Indeed, quite often univariate seasonal time series models include significant and important negative MA components

[4] A more detailed set of tables and results are reported in a companion discussion paper, see Ghysels, Lee, and Noh (1991). Each experiment is replicated 10,000 times to create sampling distributions for the test statistics. To remove the effect of the initial condition, the data are generated by setting u_{-32} and x_{-32} equal to zero, creating $T + 32$ observations, and then discarding the first 32 observations. The simulations were performed using the RATS Version 3.11 computer program.

at seasonal lags such as θ_4 in (3.5) [see e.g., Bell and Hillmer (1984) and Hillmer, Bell, and Tiao (1983)]. This fact is quite important *since it implies that serious size distortions in small samples, or even relatively large samples, may make the task of unit root testing at seasonal frequencies quite difficult.* Tables 1 through 3 report the empirical size of the different testing procedures at 5% significance level for the mixed AR and MA processes in (3.4) with the five different combinations of MA parameter values of $\theta_i = -0.9$, 0.0, 0.9 with $i = 1,4$ in (3.5). The size properties of the DHF and HEGY F_{1234}-tests and of the DF and HEGY t_1-tests are compared in tables 1 and 2, respectively. Table 3 reports the estimated sizes of the HEGY tests for seasonal unit roots at frequencies π and $\pi/2$: t_2, t_3, and F_{34}.

As shown in table 1, the size estimates based on the AR approximations are biased by the presence of MA components in the DGP. When $|\theta_1|$ is close to unity [i.e., $\theta_1 = 0.9$ or -0.9], the HEGY F_{1234}-tests appear to be severely affected by the presence of MA components, although the bias shrinks as additional lags of AR terms are included in the model. The empirical size of the DHF tests, on the other hand, seems relatively close to the nominal level. The differences in size properties of the F_{1234} and DHF for $|\theta_1|$ close to unity can be explained via the near cancellation problem in the AR and MA components [i.e., $(1 + B)$ and $(1 - B)$ in the AR component versus $(1 + 0.9B)$ and $(1 - 0.9B)$ in the MA component, respectively] observed, for instance, by Dickey, Bell, and Miller (1986) and Said (1991). The size of F_{1234} is much more affected since it can be viewed as the sum of squared t-statistics. With $|\theta_1|$ being close to unity, there is a serious size distortion affecting one of the t-statistics underlying F_{1234} which might explain why the DHF is closer to its nominal size than the F_{1234}-test. The size property of a testing procedure should, however, be evaluated in connection with the power of the test. As will be discussed later [see fig. 3], the DHF tests turn out to have very low power against the time series processes generated by (3.7) with $\alpha = 1$. The power estimates are, in some cases, quite close to the nominal size of the tests. Hence, the behavior of the DHF tests in this case results from the fact that the tests do not separate unit roots at different frequencies. Specifically, for example, for $\theta_1 = -0.9$, while the zero frequency unit root is hard to detect, in practice, due to the cancellation effect, the presence of seasonal unit roots at other frequencies render the estimated sizes of the DHF tests close to the nominal level.[5] When $\theta_4 = -0.9$, the sizes of both the DHF and HEGY tests from the simple AR regression model with $l = 1$ severely overestimate the nominal level of the tests, although the distortion becomes significantly smaller when additional lags of AR terms are included in the regression model.

[5] Similar arguments can be made for the case when $\theta_1 = 0.9$; namely, the unit root at frequency π is hard to detect in this case, but unit roots at frequencies 0 and $\pi/2$ might induce the size estimates to appear close to the nominal size of the tests.

Table 1

Empirical size of DHF and HEGY F_{1234}-tests at the 5% level with SARIMA models.

DGP: $x_t = x_{t-4} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_4 \varepsilon_{t-4}$

	AR expansion		DHF normalized bias				DHF t-statistic				HEGY F_{1234}			
			$l=1$	$l=3$	$l=4$	$l=8$	$l=1$	$l=3$	$l=4$	$l=8$	$l=1$	$l=3$	$l=4$	$l=8$
	θ_1	θ_4	*Regression with a constant*											
$T = 100$	−0.9	0.0	0.01	0.01	0.01	0.01	0.03	0.04	0.02	0.02	1.00	0.77	0.56	0.18
	0.0	−0.9	1.00	1.00	1.00	0.99	1.00	1.00	0.99	0.97	1.00	1.00	1.00	0.88
	0.0	0.0	0.03	0.03	0.02	0.03	0.04	0.05	0.04	0.04	0.05	0.05	0.04	0.05
	0.9	0.0	0.05	0.04	0.03	0.02	0.09	0.13	0.06	0.05	0.84	0.45	0.27	0.10
	0.0	0.9	0.00	0.00	0.01	0.00	0.01	0.01	0.10	0.02	0.11	0.15	0.11	0.05
$T = 400$	−0.9	0.0	0.02	0.02	0.01	0.01	0.03	0.05	0.02	0.02	1.00	0.93	0.83	0.41
	0.0	−0.9	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	0.0	0.0	0.04	0.04	0.05	0.04	0.05	0.04	0.05	0.04	0.06	0.05	0.05	0.05
	0.9	0.0	0.06	0.07	0.03	0.02	0.09	0.16	0.07	0.06	0.94	0.65	0.48	0.19
	0.0	0.9	0.00	0.00	0.02	0.02	0.01	0.01	0.12	0.02	0.09	0.09	0.12	0.04
	θ_1	θ_4	*Regression with a constant and seasonal dummies*											
$T = 100$	−0.9	0.0	0.03	0.03	0.01	0.02	0.05	0.06	0.02	0.01	0.96	0.61	0.38	0.11
	0.0	−0.9	1.00	1.00	1.00	1.00	1.00	1.00	0.95	0.95	1.00	1.00	1.00	0.89
	0.0	0.0	0.05	0.05	0.06	0.07	0.05	0.05	0.04	0.04	0.04	0.04	0.04	0.03
	0.9	0.0	0.03	0.03	0.02	0.02	0.05	0.06	0.02	0.01	0.96	0.60	0.39	0.12
	0.0	0.9	0.00	0.00	0.01	0.00	0.02	0.02	0.15	0.02	0.04	0.04	0.14	0.03
$T = 400$	−0.9	0.0	0.03	0.02	0.01	0.01	0.05	0.06	0.02	0.01	1.00	0.87	0.72	0.29
	0.0	−0.9	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	0.0	0.0	0.05	0.05	0.05	0.05	0.05	0.05	0.05	0.04	0.05	0.05	0.05	0.05
	0.9	0.0	0.03	0.03	0.01	0.01	0.05	0.07	0.01	0.01	1.00	0.87	0.72	0.28
	0.0	0.9	0.00	0.00	0.01	0.00	0.01	0.01	0.15	0.02	0.03	0.03	0.14	0.02

Table 2

Empirical size of DF and HEGY t_1-tests at the 5% level with SARIMA models.

DGP: $x_t = x_{t-4} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_4 \varepsilon_{t-4}$

	AR expansion		DF normalized bias				DF t-statistic				HEGY t_1-statistic			
			$l=0$	$l=1$	$l=3$	$l=4$	$l=0$	$l=1$	$l=3$	$l=4$	$l=0$	$l=1$	$l=3$	$l=4$
	θ_1	θ_4	*Regression with a constant*											
$T=100$	−0.9	0.0	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.86	0.69
	0.0	−0.9	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.98	1.00	0.98	0.95	0.74
	0.0	0.0	1.00	1.00	0.59	0.59	1.00	1.00	0.05	0.05	0.05	0.05	0.05	0.04
	0.9	0.0	1.00	1.00	0.30	0.38	1.00	1.00	0.03	0.08	0.03	0.08	0.06	0.04
	0.0	0.9	1.00	1.00	0.35	0.33	1.00	1.00	0.03	0.02	0.03	0.03	0.03	0.11
$T=400$	−0.9	0.0	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.96	0.89
	0.0	−0.9	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	0.0	0.0	1.00	1.00	0.60	0.60	1.00	1.00	0.05	0.04	0.05	0.05	0.05	0.05
	0.9	0.0	1.00	1.00	0.31	0.41	1.00	1.00	0.03	0.10	0.03	0.10	0.07	0.03
	0.0	0.9	1.00	1.00	1.00	0.32	1.00	1.00	0.02	0.02	0.02	0.02	0.03	0.11
	θ_1	θ_4	*Regression with a constant and a trend*											
$T=100$	−0.9	0.0	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.92	0.75
	0.0	−0.9	1.00	1.00	1.00	1.00	1.00	1.00	0.99	0.95	1.00	0.95	0.85	0.51
	0.0	0.0	1.00	1.00	0.80	0.80	1.00	1.00	0.04	0.04	0.04	0.04	0.04	0.04
	0.9	0.0	1.00	1.00	0.41	0.56	1.00	1.00	0.02	0.10	0.02	0.10	0.01	0.03
	0.0	0.9	1.00	1.00	0.50	0.47	1.00	1.00	0.01	0.01	0.02	0.01	0.01	0.13
$T=400$	−0.9	0.0	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.99
	0.0	−0.9	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	0.0	0.0	1.00	1.00	0.79	0.79	1.00	1.00	0.05	0.05	0.04	0.04	0.05	0.04
	0.9	0.0	1.00	1.00	0.40	0.59	1.00	1.00	0.00	0.13	0.01	0.13	0.09	0.02
	0.0	0.9	1.00	1.00	0.44	0.45	1.00	1.00	0.01	0.01	0.01	0.01	0.01	0.15

The main conclusion to be drawn from table 1 is that serious size distortions are an obstacle to the application of unit root tests at seasonal frequencies. Neither DHF nor HEGY handle the case $\theta_4 = -0.9$ adequately, even with $T = 400$. As negative MA components are frequently encountered in practice, this poses a very serious challenge to the design of tests for unit roots at seasonal frequencies.

Table 2 reports the size estimates of the DF unit root tests and of its HEGY counterpart t_1. The behavior of the DF t-test first indicates that the test statistics from the AR regression models of the form (3.3) with $l < 3$ are seriously biased. Comparing the results with the HEGY test, however, we can see that the distribution of the DF t-statistic with $(l + 3)$ lags is exactly the same as that of the HEGY t_1-statistic with l lags. The normalized-bias statistic, on the other hand, significantly overestimates the nominal size of the tests, which is expected from the discussion in the previous section.[6] This observation indicates that, while special care must be taken regarding the AR polynomial expansion, the usual DF-type regression model can still be employed, aside from the need to modify the normalized-bias statistic, to test for a unit root at the zero frequency even in the presence of seasonal unit roots. As discussed above, we can observe the near cancellation effect in the AR and MA components when $\theta_1 = -0.9$. This, in fact, confirms some of the observations made regarding the test statistic F_{1234}.

The size distortions reported in table 2 regarding the use of the DF test in seasonal time series with negative seasonal MA roots close to unity raises again a serious problem. Indeed, when testing for a unit root at the zero frequency, one can either (1) seasonally adjust the series and apply DF tests or (2) apply DF directly to the unadjusted series with the appropriate lag augmentation. The first strategy does not entail any major size distortions, but seasonal adjustment filters, like the commonly used X-11 filter, result in substantially reduced power for DF tests, as noted and discussed in detail in Ghysels and Perron (1993). On the other hand, the second strategy, as revealed by table 2, results most often in serious size distortions. Clearly, neither strategy is particularly satisfactory for testing zero frequency unit roots in seasonal time series.

Other aspects of the HEGY procedure can be inferred from table 3, which presents the empirical size of the HEGY tests for seasonal unit roots: t_2, t_3, and F_{34}. First, the sequential t-test for π_4 and π_3, suggested by HEGY (1990), tends to overestimate the nominal size of the tests. This result might reflect the apparent multiple testing problem. That is, since we are testing two coefficients, π_3 and π_4, with each test having 5% rejection probability under the null hypothesis, there would clearly be a greater chance of at least one rejection than the chosen 5% level. On the other hand, the empirical size of the F_{34}-test is

[6] As noted in (2.5), the normalized-bias statistic should be divided by four to be compared with the critical values in DF (1979).

Table 3

Empirical size of HEGY t_2-, t_3-, F_{34}-tests at the 5% level with SARIMA models.

DGP: $x_t = x_{t-4} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_4 \varepsilon_{t-4}$

	AR expansion		HEGY t_2				HEGY t_3				HEGY F_{34}			
			$l=1$	$l=3$	$l=4$	$l=8$	$l=1$	$l=3$	$l=4$	$l=8$	$l=1$	$l=3$	$l=4$	$l=8$
	θ_1	θ_4	*Regression with a constant*											
$T=100$	−0.9	0.0	0.06	0.05	0.04	0.04	0.03	0.05	0.03	0.03	0.05	0.05	0.04	0.05
	0.0	−0.9	0.98	0.94	0.84	0.58	1.00	1.00	0.98	0.86	1.00	0.99	0.94	0.70
	0.0	0.0	0.05	0.04	0.04	0.04	0.10	0.10	0.08	0.09	0.05	0.05	0.04	0.05
	0.9	0.0	0.94	0.67	0.55	0.14	0.03	0.06	0.03	0.03	0.07	0.05	0.04	0.05
	0.0	0.9	0.02	0.01	0.09	0.03	0.17	0.15	0.12	0.10	0.18	0.15	0.06	0.08
$T=400$	−0.9	0.0	0.09	0.07	0.04	0.04	0.03	0.07	0.03	0.04	0.08	0.06	0.06	0.05
	0.0	−0.9	1.00	1.00	1.00	0.90	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.96
	0.0	0.0	0.05	0.05	0.05	0.05	0.10	0.09	0.10	0.09	0.04	0.06	0.06	0.05
	0.9	0.0	0.99	0.82	0.70	0.39	0.01	0.08	0.03	0.03	0.10	0.06	0.06	0.05
	0.0	0.9	0.01	0.01	0.11	0.03	0.16	0.15	0.14	0.10	0.17	0.16	0.06	0.07
	θ_1	θ_4	*Regression with a constant and a trend*											
$T=100$	−0.9	0.0	0.05	0.04	0.03	0.03	0.02	0.02	0.02	0.01	0.05	0.05	0.05	0.08
	0.0	−0.9	0.98	0.94	0.85	0.58	1.00	1.00	0.98	0.86	1.00	0.99	0.95	0.71
	0.0	0.0	0.03	0.03	0.03	0.03	0.09	0.08	0.07	0.08	0.05	0.05	0.04	0.04
	0.9	0.0	1.00	0.82	0.65	0.20	0.02	0.03	0.01	0.02	0.05	0.06	0.05	0.04
	0.0	0.9	0.02	0.01	0.09	0.03	0.17	0.15	0.03	0.02	0.18	0.16	0.07	0.08
$T=400$	−0.9	0.0	0.08	0.06	0.04	0.04	0.02	0.04	0.01	0.02	0.05	0.07	0.06	0.06
	0.0	−0.9	1.00	1.00	1.00	0.90	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.96
	0.0	0.0	0.05	0.04	0.05	0.04	0.09	0.09	0.09	0.09	0.05	0.05	0.05	0.05
	0.9	0.0	1.00	0.95	0.88	0.48	0.02	0.05	0.02	0.02	0.05	0.07	0.06	0.06
	0.0	0.9	0.01	0.01	0.11	0.03	0.16	0.14	0.14	0.04	0.17	0.17	0.07	0.07

$$\textbf{Model} : \mathbf{x}_t = x_{t-4} + \sum_{i=1}^{4} \delta_i SD_{it} + \mu_t$$

SD1 : $\delta_1 = -0.1; \delta_2 = 0.05; \delta_3 = -0.05; \delta_4 = 0.1$ SD2 : $\delta_1 = -1.0; \delta_2 = 0.5; \delta_3 = -0.5; \delta_4 = 1.0$

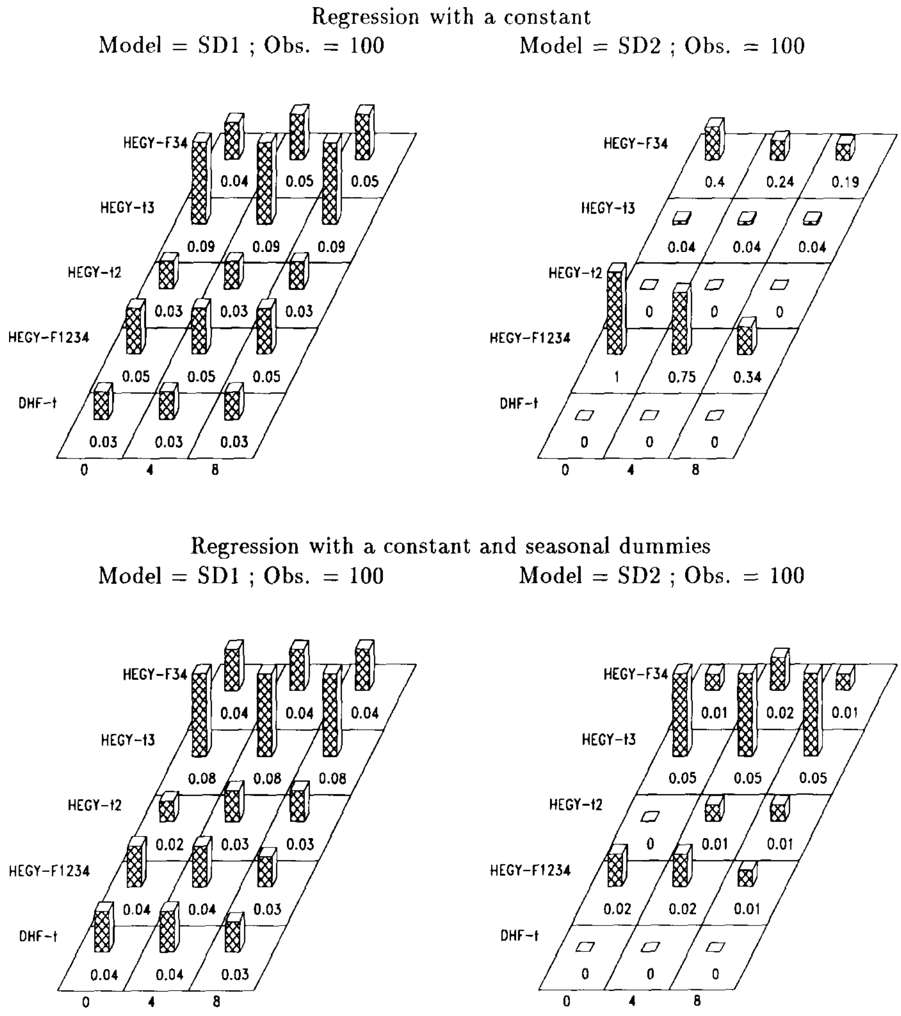

Fig. 1. Empirical size of DHF and HEGY F_{1234}-, t_2-, t_3-, and F_{34}-tests at the 5% level for SARIMA (p, d, 0) models with seasonal dummies.

relatively close to the nominal level, suggesting that this test should be preferred in testing for unit roots at frequency $\pi/2$. We can again observe the near cancellation phenomenon for $\theta_1 = 0.9$ between $(1 + B)$ in the AR components and $(1 + \theta_1 B)$ in the MA components, which induces a significant bias of the t_2-statistic.

The size properties of the tests for the time series with mixed deterministic and stochastic seasonality generated by (3.6) are considered with different configurations of the seasonal dummy coefficients. As shown in fig. 1, the size of the tests is affected by the presence of seasonal dummies, particularly when the regression models without seasonal dummies are used (the larger the variation in the seasonal dummy coefficients, the larger the bias). Specifically, the regression models with seasonal dummies appear to provide reasonable estimates of the size, while the estimated sizes of the tests from the regressions without seasonal dummies are seriously biased. This result indicates that the test statistics without seasonal dummy variables do not have the distributions in DHF (1984) and HEGY (1990), but are functions of the unknown seasonal dummy coefficients, δ_i's $(i = 1,2,3,4)$ [see Evans and Savin (1987)]. Similar observations can be drawn for the empirical size of the HEGY tests for seasonal unit roots in the DGP (3.6). The outcome of the tests are also quite sensitive to the inclusion of seasonal dummies in the regression models. Moreover, the multiple testing problem of the sequential t-test for π_4 and π_3 also appears in this case as it did in table 3.

Since it is well known that the properties at the HEGY t_1-statistic are similar to those of the t_2-statistic when a constant and seasonal dummies are included in the regression model, the results for the HEGY t_1-test together with those for the DF test are suppressed.[7] Given the similarity between the HEGY t_1-test and t_2-test, however, it is worth noting that the t_1-test without a constant term tends to be seriously affected by the presence of seasonal dummies, as the estimated sizes of the t_2-test without seasonal dummies are seriously biased. In fact, the t_1-statistic without a constant term does not have the limiting distribution in DF (1979), but is also a function of the unknown seasonal dummy coefficients, δ_i's $(i = 1,2,3,4)$.

3.2.2. *Power of the tests*

There is little point of discussing power properties of tests which have severe size distortions, as uncorrected defective tests hardly seem appropriate. Consequently, our investigation of power properties focuses on cases reported in the previous section, where no major size distortions occur. This means that we

[7] The properties of the DF test in comparison with the HEGY test discussed above also apply in this case. That is, the behavior of the DF t-statistic with $(l + 3)$ lags is the same as its HEGY counterpart, t_1, with l lags, while the normalized-bias statistic, $T(\hat{\phi}_1 - 0)$, overestimates the nominal size of the test.

analyze the power properties for DGP's described by (3.6) and (3.7) involving deterministic seasonal mean shifts.

The power of the tests for the time series with mixed deterministic and stochastic seasonality generated by (3.6) are considered using the AR parameter $\alpha = 0.85$. As shown in fig. 2, the regressions without seasonal dummies have very low power against the DGP with seasonal dummies; the larger the seasonal variation, the lower the power of the tests. The regressions with seasonal dummies, on the other hand, have the usual power against this type of time series processes. Hence, based on the simulation results, inclusion of a constant and seasonal dummies appears to be a prudent decision in empirical applications in order to perform tests for seasonal unit roots. As expected, we note that the power of the test increases as T grows. This observation is even clearer when we look at the power properties of the HEGY tests for seasonal unit roots. Another interesting point to note here is that the power of the tests at frequency $\pi/2$ compares favorably with that of the tests at frequency π. This observation might be valid only for the processes of a particular type such as those generated by (3.6) as, in this case, the coefficient π_4 is zero both under the null and alternative hypotheses. In a more general case where $\pi_4 \neq 0$ under the alternative, the power of the tests for seasonal unit roots at different frequencies might be close to each other. Since the properties of the HEGY t_1-statistic are similar to those of the t_2-statistic if a constant and seasonal dummies are included in the model, the result for the HEGY t_1-test together with that for the DF test is again suppressed in this case.

Since the time series generated by (3.7) contain only deterministic seasonal components, no seasonal unit roots are present in the time series processes of this type. Hence, the behavior of the DHF procedure, which jointly tests for the presence of all of the four unit roots, will show the power properties of the test. The powers of the DHF and its HEGY counterpart F_{1234}-tests against the DGP in (3.7) with $\alpha = 1$ are compared in fig. 3, which shows that the power of the DHF tests against this type of process is very low, relative to the HEGY procedure.[8] On the other hand, the HEGY tests generally perform very well in this case. This result implies that the DHF test may not separate unit roots at each frequency, while the HEGY F_{1234}-test, although a joint test, does in fact appear to test for unit roots at each frequency separately (see also table 1).

The above comparison between the DHF and HEGY tests also applies in the case when $\alpha = 0.9$. That is, while the DHF test appears to perform relatively well compared to the case when $\alpha = 1$, the power of the DHF test is generally lower than that of its HEGY counterpart. As is the case with the usual tests for a unit root at the zero frequency, a comparison between panels A and B shows that the

[8] As the power estimates in our simulations in this case did not appear to depend much on the sample sizes considered and the magnitude of seasonal dummy coefficients, δ_i's, the results for $T = 100$ and $((\delta_i), i = 1, \dots, 4) = (-1, 0.5, -0.5, 1)$ are only reported here.

$$\textbf{Model}: x_t = 0.85x_{t-4} + \sum_{i=1}^{4} \delta_i SD_{it} + \mu_t$$

SD1 : $\delta_1 = -0.1; \delta_2 = 0.05; \delta_3 = -0.05; \delta_4 = 0.1$ SD2 : $\delta_1 = -1.0; \delta_2 = 0.5; \delta_3 = -0.5; \delta_4 = 1.0$

Regression with a constant

Model = SD1 , Obs. = 100 Model = SD2 , Obs. = 100

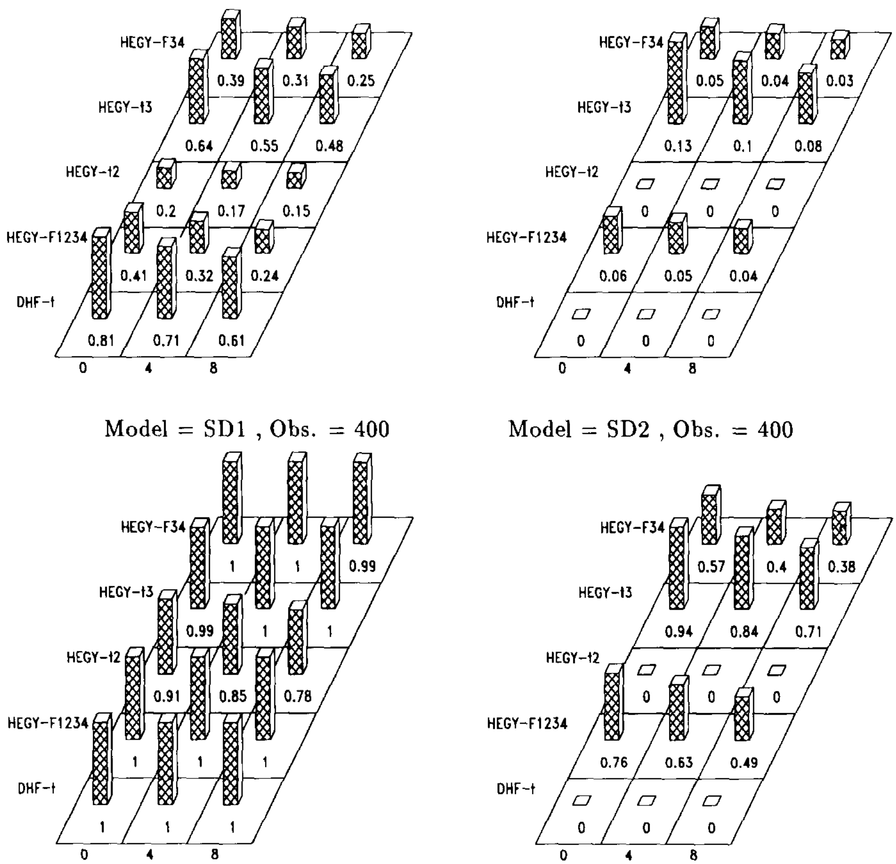

Fig. 2. Empirical power of DHF and HEGY F_{1234}-, t_2-, t_3-, and F_{34}-tests at the 5% level for SARIMA (p, d, 0) models with seasonal dummies.

power increases as α gets smaller. Finally, we consider the power of the HEGY t_2-, t_3-, and F_{34}-tests against alternatives drawn from (3.7) with $\alpha = 0.9$. Hence, there are no unit roots present anywhere, i.e., at zero as well as seasonal

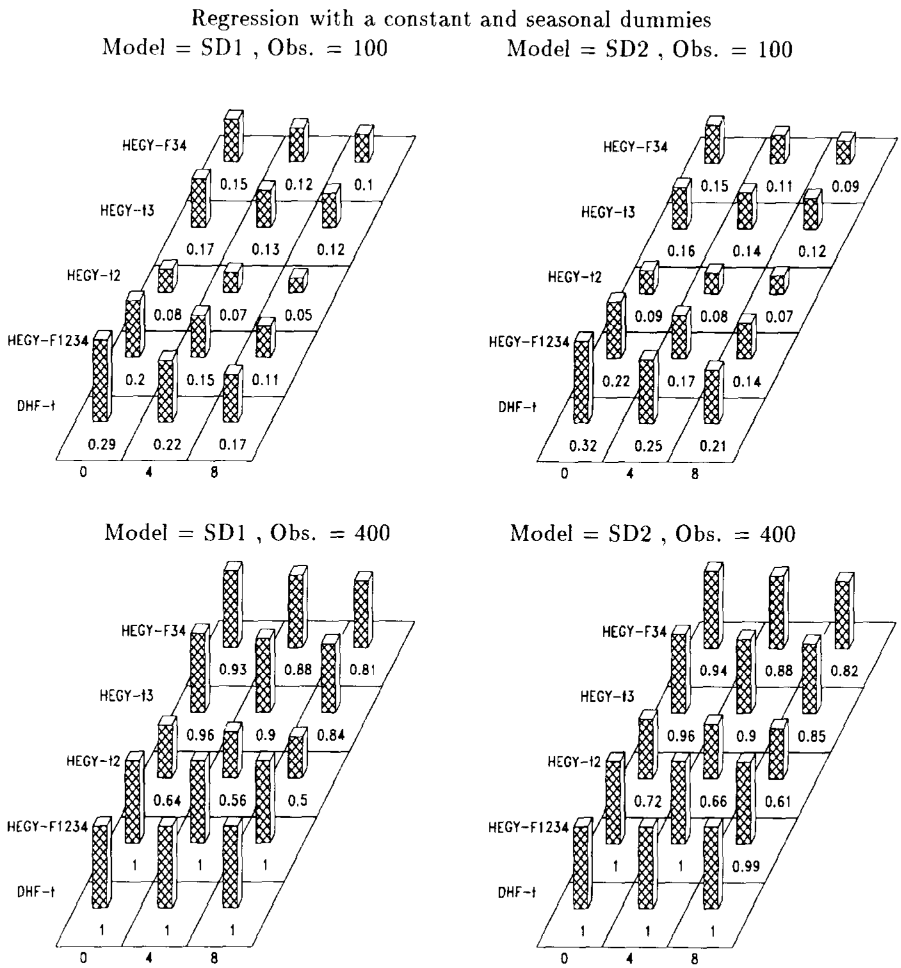

Fig. 2 (continued)

frequencies. From fig. 3, we observe that the HEGY t_2, t_3, and F_{34} are quite powerful. This is an encouraging result since the DGP in panel C is a plausible alternative in many practical situations.

4. Concluding remarks

For the practical researcher, the following should emerge as the main conclusions from our theoretical extensions and Monte Carlo investigation. First, as the HEGY procedure can test for the presence of unit roots at each frequency separately without assuming that other unit roots are present,

$$\textbf{Model}: x_t = \alpha x_{t-1} + \sum_{i=1}^{4} \delta_i SD_{it} + \mu_t \quad ; \delta_1 = -1.0; \delta_2 = 0.5; \delta_3 = -0.5; \delta_4 = 1.0$$

Regression with a constant — Regression with a constant and seasonal dummies

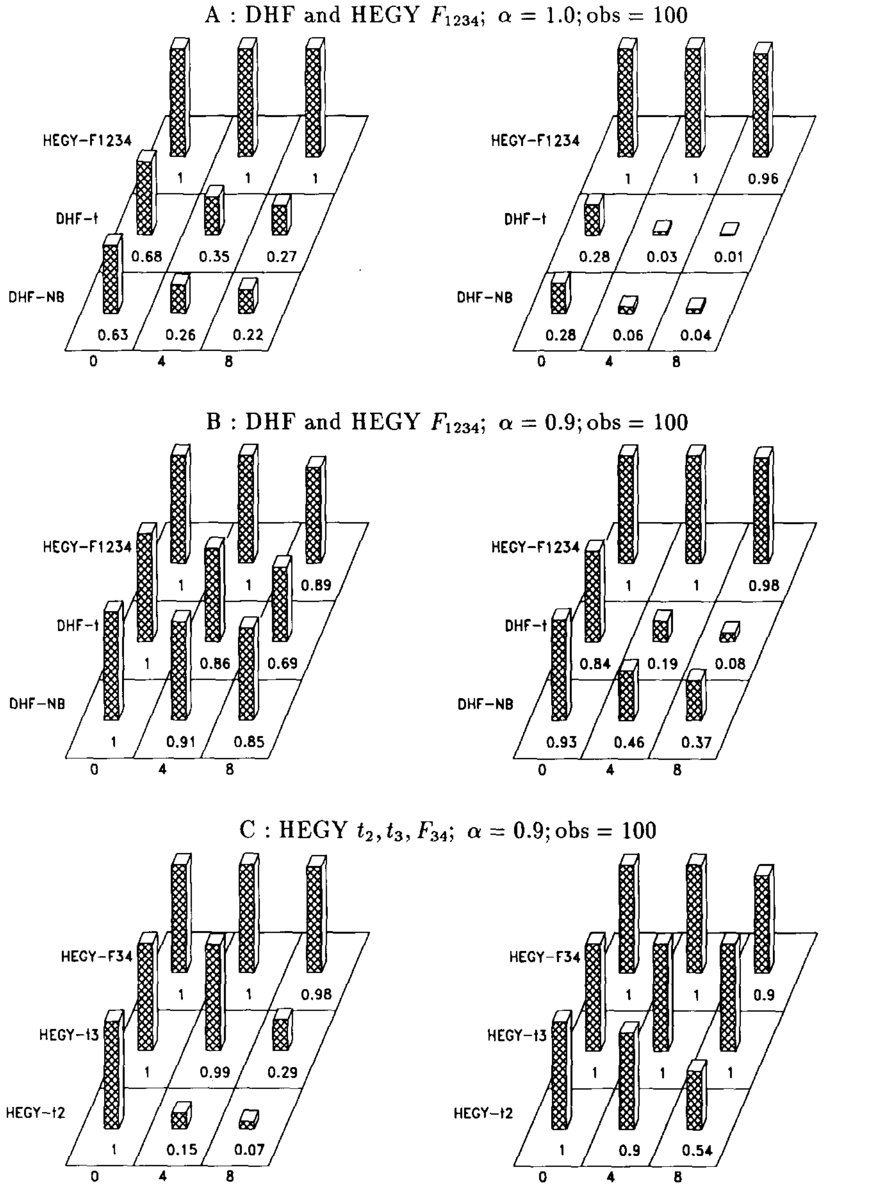

Fig. 3. Empirical power of DHF and HEGY F_{1234}-, t_2-, t_3-, and F_{34}-tests at the 5% level for ARI (p, d) models with seasonal dummies.

it is expected to be most useful among the alternative testing procedures like DHF in detecting certain types of nonstationarity. Second, while HEGY has a design which discriminates more the presence of roots at the various frequencies on the unit circle, it has the drawback of being more sensitive to size distortions due to near-cancellations of AR unit roots with MA polynomial roots at the different frequencies. In practice, in seasonal time series models, one often encounters MA coefficients in the near-cancellation region. Hence, size distortions are believed to be potentially quite serious when applying HEGY-type tests in moderate or even large samples. Third, it was found that when the data-generating processes have seasonal dummies, the regression without seasonal dummies seriously distorts the test result [i.e., it leads to a large bias in the size or too low power]. Hence, although inclusion of too many lags or irrelevant deterministic terms (i.e., a constant, seasonal dummies, and/or a trend) tends to reduce the power of the tests, the safe strategy in empirical applications is the inclusion of these (possibly irrelevant) terms in the model.

Overall, the picture drawn from our investigation is not very encouraging. Tests currently available suffer from a combination of two serious shortcomings, as already noted, namely (1) size distortions in the most practically relevant cases and (2) in situations where the size is adequate, low power against plausible alternatives.

Another important result of practical use is the observation that one can still apply the usual DF test for a zero frequency unit root even in the presence of unit roots at some of the seasonal frequencies, provided appropriate autoregressive correction terms are included in the regression model. But, once again, there are serious size distortions. This faces the practical researcher with a difficult choice. Namely, either using unadjusted data resulting in tests with the wrong size, or using adjusted data, with adjustment procedures having adverse effects on power, as discussed by Ghysels and Perron (1993).

Finally, while the Monte Carlo experiment in the previous section covers many interesting cases in seasonal time series models, it does not include all models that might be fit for seasonal data. One case which might arise is where the seasonal dummy coefficients are changing over time as discussed by Ghysels (1991). This case seems to parallel the discussion about the effect of structural breaks on unit root tests in Hendry and Neale (1989) and Perron (1989). We may indeed tend to find unit roots at seasonal frequencies because of changes in seasonal patterns unaccounted for in tests like HEGY, DHF, and many others.

Appendix A

We first derive the distribution of F_{1234} appearing in (2.3). Note that

$$F_{1234} = \frac{\hat{\pi}'[\sum Y_t Y_t']\hat{\pi}/4}{\sum \hat{e}_t^2/(T-4)},$$

where $\hat{\pi} = (\hat{\pi}_1\ \hat{\pi}_2\ \hat{\pi}_3\ \hat{\pi}_4)'$ and $Y_t = (y_{1,t-1}\ y_{2,t-1}\ y_{3,t-2}\ y_{3,t-1})'$. Using the results on the limiting distribution of $T\hat{\pi}_j\,(j = 1,2,3,4)$ and $T^{-2}\sum y_{j,t}^2\,(j = 1,2,3)$ [see Theorem 3.5.1 of Chan and Wei (1988) and Engle, Granger, Hylleberg, and Lee (1993)], we have

$$\hat{\pi}'(\textstyle\sum Y_t Y_t')\hat{\pi} = \hat{\pi}_1^2 \sum y_{1,t-1}^2 + \hat{\pi}_2^2 \sum y_{2,t-1}^2 + \hat{\pi}_3^2 \sum y_{3,t-2}^2 + \hat{\pi}_4^2 \sum y_{3,t-1}^2 + o_p(0).$$

The asymptotic distribution of the F-test on $\pi_1 = \pi_2 = \pi_3 = \pi_4 = 0$ can now be derived by using the relation

$$\textstyle\sum \hat{e}_t^2/(T-4) \to \sigma_\varepsilon^2,$$

which establishes the result in (2.3). Next, we turn to the distribution of F_{234}. Noting that

$$F_{234} = \frac{(R\hat{\pi})'[R(\sum Y_t Y_t')^{-1}R']^{-1}(R\hat{\pi})/3}{\sum \hat{e}_t^2/(T-4)},$$

where

$$R = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

the limiting distribution of F-statistic for $\pi_2 = \pi_3 = \pi_4 = 0$ can similarly be derived and the details omitted.

Appendix B

We consider an autoregressive model of order p,

$$x_t = \phi_1 x_{t-1} + \phi_2 x_{t-2} + \cdots + \phi_p x_{t-p} + \varepsilon_t, \qquad t = 1, 2, \ldots, T, \tag{A.1}$$

where ε_t is an arbitrary innovation process. Using the Proposition in HEGY (1990), we can first rewrite eq. (A.1) as

$$\Delta x_t = \alpha_1 x_{t-1} + \alpha_2 \Delta x_{t-1} + \alpha_3 \Delta x_{t-2} + \alpha_4 \Delta x_{t-3} + \sum_{i=1}^{p-4} \alpha_{4+i} \Delta x_{t-3-i} + \varepsilon_t, \tag{A.2}$$

$$\alpha_1 = 1 - \phi_1 - \phi_2 - \cdots - \phi_p,$$

$$\alpha_2 = -\phi_2 - \phi_3 - \cdots\cdots - \phi_p,$$

$$\vdots$$

$$\alpha_{p-1} = -\phi_{p-1} - \phi_p,$$

$$\alpha_p = -\phi_p.$$

We can alternatively reparameterize (A.1) as follows:

$$\Delta_4 x_t = \pi_1 y_{1,t-1} + \pi_2 y_{2,t-1} + \pi_3 y_{3,t-2} + \pi_4 y_{3,t-1} + \sum_{i=1}^{p-4} \pi_{4+i} \Delta_4 x_{t-i} + \varepsilon_t, \tag{A.3}$$

where the y_{jt}'s ($j = 1, 2, 3$) are defined in (2.2).

For notational convenience, we shall use the following symbols:

$$\alpha = (\alpha_1\ \alpha_2\ \alpha_3\ \alpha_4 \ldots \alpha_p)',$$

$$Y_{\alpha t} = (x_{t-1}\ \Delta x_{t-1}\ \Delta x_{t-2}\ \Delta x_{t-3}\ \Delta x_{t-4}\ \cdots\ \Delta x_{t-p+1})',$$

$$Y_{\pi t} = (y_{1,t-1}\ y_{2,t-1}\ y_{3,t-2}\ y_{3,t-1}\ \Delta_4 x_{t-1}\ \cdots\ \Delta_4 x_{t-p+4})'.$$

We shall also use a $p \times p$ matrix Q of the form [see Chan and Wei (1988)]:

$$\left[\begin{array}{cccc|ccccccc}
4 & -3 & -2 & -1 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 \\
0 & -1 & 0 & -1 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 \\
0 & 0 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 \\
0 & -1 & -1 & 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 0 \\
\hline
0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & \cdots & 0 \\
0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 & \cdots & 0 \\
0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & \cdots & 0 \\
0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & \cdots & 0 \\
\vdots & \vdots & \vdots & \vdots & & & & & & & \\
0 & 0 & 0 & 0 & 0 & \cdots & 0 & 1 & 1 & 1 & 1
\end{array}\right]$$

We first derive the relations (2.5) and (2.6) for a general AR model of the form in (A.1) under the assumption that ε_t follows a martingale difference sequence obeying the conditions for the functional central limit theorem in Phillips (1987). Using Theorem 3.5.1 of Chan and Wei (1988), we have for the first element of a $p \times 1$ vector $(Q')^{-1}(\hat{\alpha} - \alpha)$ that

$$
\begin{aligned}
T[(Q')^{-1}(\hat{\alpha} - \alpha)]_{11} &= T[(Q')^{-1}(\textstyle\sum Y_{\alpha t} Y'_{\alpha t})^{-1}(\textstyle\sum Y_{\alpha t}\varepsilon_t)]_{11} \\
&= T(\textstyle\sum y_{1,t-1}^2)^{-1}(\textstyle\sum y_{1,t-1}\varepsilon_t) + \mathrm{o}_p(0) \\
&\to \{W(1)^2 - 1\}/2\int_0^1 W(r)^2\,\mathrm{d}r.
\end{aligned}
$$

The relation (2.5) follows by noting that

$$
(Q')^{-1} = \begin{bmatrix} \frac{1}{4} & 0 & 0 & 0 & 0 & \cdots & 0 \\ * & * & * & * & * & \cdots & * \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdots & \cdot \\ \cdot & \cdot & \cdot & \cdot & \cdot & \cdots & \cdot \\ * & * & * & * & * & \cdots & * \end{bmatrix},
$$

where $*$ denotes elements of the matrix $(Q')^{-1}$ which might have values different from zero. The expression (2.6) is also immediate from the following relation:

$$
\begin{aligned}
T^{-2}Q(\textstyle\sum Y_{\alpha t} Y'_{\alpha t})Q' &= T^{-2}\textstyle\sum y_{1,t-1}^2 + \mathrm{o}_p(0) \\
&\to \int_0^1 W(r)^2\,\mathrm{d}r.
\end{aligned}
$$

We now show that the distribution of the t-ratio on π_1 in the HEGY regression (A.3) is the same as that on α_1 in the DF regression (A.2). Denoting the first element of a matrix [ ] by $[\ ]_{11}$, we have

$$
\begin{aligned}
t\hat{\alpha} &= [\hat{\sigma}_\varepsilon^2(\textstyle\sum Y_{\alpha t} Y'_{\alpha t})^{-1}]_{11}^{-1/2}[(\textstyle\sum Y_{\alpha t} Y'_{\alpha t})^{-1}(\textstyle\sum Y_{\alpha t}\varepsilon_t)]_{11} \\
&= 1/\hat{\sigma}_\varepsilon[Q'(\textstyle\sum QY_{\alpha t} Y'_{\alpha t}Q')^{-1}Q]_{11}^{-1/2}[Q'(\textstyle\sum QY_{\alpha t} Y'_{\alpha t}Q')^{-1}QQ^{-1}(\textstyle\sum QY_{\alpha t}\varepsilon_t)]_{11} \\
&= 1/\hat{\sigma}_\varepsilon[Q'(\textstyle\sum Y_{\pi t} Y'_{\pi t})^{-1}Q]_{11}^{-1/2}[Q'(\textstyle\sum Y_{\pi t} Y'_{\pi t})^{-1}(\textstyle\sum Y_{\pi t}\varepsilon_t)]_{11} \\
&= 1/\hat{\sigma}_\varepsilon[16(\textstyle\sum Y_{\pi t} Y'_{\pi t})^{-1}]_{11}^{-1/2}[4(\textstyle\sum Y_{\pi t} Y'_{\pi t})^{-1}(\textstyle\sum Y_{\pi t}\varepsilon_t)]_{11} \\
&= [\hat{\sigma}_\varepsilon^2(\textstyle\sum Y_{\pi t} Y'_{\pi t})^{-1}]_{11}^{-1/2}[(\textstyle\sum Y_{\pi t} Y'_{\pi t})^{-1}(\textstyle\sum Y_{\pi t}\varepsilon_t)]_{11} \\
&= t_1\,.
\end{aligned}
$$

Since the limiting distribution of the HEGY t_1-statistic has been shown to follow the distribution in DF (1979), the above relation also proves the expression in (2.6).

Appendix C

Table C.1

Critical values of F_{1234} and F_{234} test statistics.[a]

Testing procedure	$T = 48$	$T = 100$	$T = 160$	$T = 200$	$T = 400$
		5% critical values			
hf24000	2.80	2.76	2.72	2.73	2.76
hf14000	2.62	2.55	2.53	2.56	2.53
hf24100	2.63	2.74	2.74	2.75	2.69
hf14100	3.47	3.37	3.38	3.32	3.29
hf24101	2.67	2.76	2.75	2.72	2.70
hf14101	4.28	4.26	4.13	4.12	4.09
hf24110	6.06	6.05	6.04	5.99	5.95
hf14110	5.96	5.74	5.69	5.58	5.55
hf24111	6.09	5.99	5.91	5.89	5.90
hf14111	6.53	6.47	6.33	6.38	6.27
		10% critical values			
hf24000	2.21	2.21	2.18	2.22	2.19
hf14000	2.11	2.11	2.06	2.12	2.07
hf24100	2.17	2.18	2.21	2.19	2.15
hf14100	2.86	2.86	2.81	2.81	2.79
hf24101	2.14	2.15	2.15	2.17	2.16
hf14101	3.60	3.58	3.55	3.55	3.54
hf24110	5.16	5.18	5.16	5.14	5.15
hf14110	5.02	4.95	4.90	4.87	4.86
hf24111	5.13	5.13	5.14	5.10	5.10
hf14111	5.71	5.68	5.56	5.61	5.55

[a] hf14*abc* stands for the F_{1234}-statistic with $a = 0$ or 1 for no constant or constant, $b = 0$ without seasonal dummies or $b = 1$ with seasonal dummies, and $c = 0$ or 1 for linear trend being absent or present. Likewise, hf24*abc* represents the F_{234}-statistic. The critical values reported in the table were obtained via Monte Carlo simulations using the random number generator of GAUSS 2.0 with 10,000 replications.

Table C.2

Critical values of test statistics for seasonal unit roots at the 5% level.

Testing procedure	$T = 100$	$T = 400$	Testing procedure	$T = 100$	$T = 400$
dhfnb00	− 8.96	− 9.12	ht1110	− 2.95	− 2.91
dhft00	− 1.89	− 1.90	ht2110	− 2.94	− 2.89
dhfnb10	− 12.22	− 12.54	ht2110	− 3.44	− 3.38
dhft10	− 2.38	− 2.38	ht4110	2.29	2.32
dhfnb11	− 25.52	− 27.32	hf34110	6.57	6.61
dhft11	− 4.10	− 4.05			

Table C.2 (continued)

Testing procedure	$T = 100$	$T = 400$	Testing procedure	$T = 100$	$T = 400$
ht1000	− 1.97	− 1.94	ht1111	− 3.53	− 3.49
ht2000	− 1.92	− 1.95	ht2111	− 2.94	− 2.91
ht3000	− 1.90	− 1.92	ht3111	− 3.48	− 3.41
ht4000	2.00	1.97	ht4111	2.28	2.31
hf34000	3.12	3.16	hf34111	6.60	6.57
ht1100	− 2.88	− 2.87	dfnb000	− 7.90	− 8.00
ht2100	− 1.95	− 1.92	dft000	− 1.95	− 1.95
ht3100	− 1.90	− 1.90	dfnb100	− 13.70	− 14.00
ht4100	1.97	1.96	dft100	− 2.89	− 2.87
hf34100	3.08	3.05	dfnb110	− 13.70	− 14.00
			dft110	− 2.89	− 2.87
ht1101	− 3.47	− 3.44	dfnb101	− 20.70	− 21.40
ht2101	− 1.94	− 1.95	dft101	− 3.45	− 3.42
ht3101	− 1.89	− 1.92	dfnb111	− 20.70	− 21.40
ht4101	1.98	1.96	dft111	− 3.45	− 3.42
hf34101	2.98	3.07			

[a] The codes *xyabc* represent the different test statistics where x = dhf for DHF, h for HEGY, and df for DF; y = nb for normalized bias, t for t-statistics in DHF and DF regressions, and t1 through t4 and f34 for HEGY procedures. Finally, $a = 0$ or 1 for a constant, $b = 0$ or 1 for seasonal dummies, and $c = 0$ or 1 for a linear trend (see also table C.1). A more complete table for samples of sizes $T = 48$, 160, and 200 included appears in Ghysels, Lee, and Noh (1991). When the exact percentiles of the test statistics were not available, we interpolated the critical values.

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